

AI Smart Hospital - Agentic RAG Project Report

Executive Summary

AI Smart Hospital uses an Agentic Retrieval-Augmented Generation architecture to provide intelligent access to hospital information. The platform connects patient records, doctor schedules, laboratory systems, insurance databases, room management systems, and hospital policies. AI agents retrieve relevant information and generate accurate responses for patients, doctors, nurses, and administrators.

Hospital Overview

AI Smart Hospital is a multi-specialty healthcare institution providing emergency care, surgery, diagnostics, telemedicine, pharmacy, intensive care, and preventive healthcare services. The hospital operates 24 hours a day and supports both outpatient and inpatient services.

Facilities

The hospital provides emergency services, ICU, NICU, operation theaters, pharmacy, blood bank, laboratory, radiology, dialysis center, ambulance services, cafeteria, ATM, waiting lounges, wheelchair support, telemedicine facilities, and patient transportation services.

Awards and Recognition

The hospital has received sample recognitions including Excellence in Patient Care Award, Digital Healthcare Innovation Award, Community Health Service Award, and Healthcare Technology Excellence Award.

Departments

Major departments include Cardiology, Neurology, Orthopedics, Oncology, Dermatology, Pediatrics, ENT, Gastroenterology, Pulmonology, Nephrology, Urology, General Medicine, General Surgery, Gynecology, and Emergency Medicine.

Doctor Directory (20 Sample Doctors)

Doctor D001 is a specialist with more than 6 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D002 is a specialist with more than 7 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D003 is a specialist with more than 8 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D004 is a specialist with more than 9 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D005 is a specialist with more than 10 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D006 is a specialist with more than 11 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D007 is a specialist with more than 12 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D008 is a specialist with more than 13 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D009 is a specialist with more than 14 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D010 is a specialist with more than 15 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D011 is a specialist with more than 16 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D012 is a specialist with more than 17 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D013 is a specialist with more than 18 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D014 is a specialist with more than 19 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D015 is a specialist with more than 20 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D016 is a specialist with more than 21 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D017 is a specialist with more than 22 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D018 is a specialist with more than 23 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D019 is a specialist with more than 24 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Doctor D020 is a specialist with more than 25 years of experience. The doctor consults patients, reviews medical history, provides treatment recommendations, and participates in emergency care. Consultation hours are available from 9:00 AM to 5:00 PM on weekdays with scheduled appointment slots.

Patient P031 has a digital health profile containing demographics, diagnosis history, allergies, medications, laboratory reports, doctor consultations, and follow-up schedules. The Agentic RAG system retrieves only authorized information when responding to queries.

Patient P047 has a digital health profile containing demographics, diagnosis history, allergies, medications, laboratory reports, doctor consultations, and follow-up schedules. The Agentic RAG system retrieves only authorized information when responding to queries.

Patient P048 has a digital health profile containing demographics, diagnosis history, allergies, medications, laboratory reports, doctor consultations, and follow-up schedules. The Agentic RAG system retrieves only authorized information when responding to queries.

Patient P049 has a digital health profile containing demographics, diagnosis history, allergies, medications, laboratory reports, doctor consultations, and follow-up schedules. The Agentic RAG system retrieves only authorized information when responding to queries.

Patient P050 has a digital health profile containing demographics, diagnosis history, allergies, medications, laboratory reports, doctor consultations, and follow-up schedules. The Agentic RAG system retrieves only authorized information when responding to queries.

Laboratory Services (30+ Tests)

CBC: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Blood Sugar: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

HbA1c: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Lipid Profile: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Liver Function Test: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Kidney Function Test: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Thyroid Profile: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Vitamin D: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Vitamin B12: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Electrolytes: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

CRP: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

ESR: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

D-Dimer: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Troponin: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

PSA: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Urine Routine: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Urine Culture: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Stool Analysis: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Dengue: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Malaria: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

HIV: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Hepatitis B: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Hepatitis C: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

COVID PCR: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Blood Culture: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Pregnancy Test: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Allergy Profile: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Coagulation Profile: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Iron Studies: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Ferritin: This laboratory investigation supports diagnosis, treatment planning, and patient monitoring.

Appointment Booking

Patients can book appointments through the website, mobile application, hospital reception, telemedicine portal, or telephone support center. Confirmation and reminders are sent automatically.

Room Management

The hospital provides AC rooms, non-AC rooms, deluxe rooms, private rooms, semi-private rooms, general wards, ICU beds, and NICU beds. Availability is tracked in real time.

Insurance Services

Insurance partners include Star Health, Niva Bupa, HDFC ERGO, ICICI Lombard, Care Health, Aditya Birla Health, and SBI Health. Dedicated coordinators help patients with approvals and cashless claims.

Parking Management

Separate parking areas are available for visitors, staff, doctors, and emergency vehicles. Security monitoring and automated fee collection are available.

Holiday Schedule

Emergency services remain operational during public holidays. Selected outpatient departments may operate with revised schedules.

Career Portal and Vacancies

The hospital regularly recruits nurses, doctors, pharmacists, laboratory technicians, AI engineers, data engineers, software developers, and administrative staff. Applications can be submitted through the HR department.

Security Systems

The facility uses CCTV monitoring, biometric access control, visitor management, role-based authorization, audit logging, and encrypted data storage.

ICU Infrastructure

The ICU contains ventilators, patient monitors, infusion pumps, syringe pumps, defibrillators, ECG systems, oxygen delivery systems, and emergency crash carts.

Agentic RAG Architecture

Planner agents analyze user intent. Retrieval agents collect data from vector databases and hospital systems. Response agents generate accurate answers with traceable sources.

Technology Stack

The solution uses Python, FastAPI, LangChain, LangGraph, PostgreSQL, ChromaDB, and modern large language models.

Future Scope

Future enhancements include multilingual support, predictive analytics, wearable integration, AI triage, and voice-enabled healthcare assistants.